Fiscal Year:	FY 2011	**Task Last Updated:**	FY 02/22/2012
PI Name:	Yagoobi, Jamal Ph.D.		
Project Title:	A Two-Phase Heat Transport Device Driven by Electric Conduction Mechanism in Micro- and Macro Scales		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	01609-2280	**Congressional District:**	3
Comments:	NOTE PI formerly at Illinois Institute of Technology; moved to Worcester Polytechnic Institute in January 2012. Info per PI (Ed., 1/17/2012)		
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	06/01/2009	**End Date:**	09/30/2012
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: Change in end date to 9/30/2012, per PI (Ed., 2/28/2013)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Didion, Jeffrey (NASA Goddard Space Flight Center) Cassel, Kevin (Illinois Institute of Technology)		
Grant/Contract No.:	NNX09AK65G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Electrically driven flows can provide significant enhancements to current and future technologies that can benefit from increased heat transfer and mass transport capacities in macro and micro scales. In particular, electrically generated flows (with and without phase change) have great potential for terrestrial and micro-gravity applications. Electrohydrodynamic (EHD) ion-drag pumping and induction pumping have been extensively studied in the past several decades with no significant, successful applications. On the contrary, EHD pumping based on conduction phenomenon has reached a stage where various applications are expected in the near future, since pressure heads on the order of kilo Pascal per electrode pair are easily obtainable. EHD conduction pumps are simple in design, lightweight, non-mechanical, and free of vibration and noise, and they allow for active control of heat transfer and mass transport. EHD conduction pumping requires minimal electric power to operate (at least three orders of magnitude less power compared to the resultant heat transport capacity). This proposed research will fundamentally investigate the electrically-driven dielectric liquid film in the presence of phase change in macro- and micro-scales based on conduction phenomenon through a combined experimental and computational investigation. Furthermore, the coupling of electric double layers and heterocharge layers in micro-channels will be explored, and the non-dimensional threshold at which the double layer starts to affect the corresponding flow field will be determined. Note that in micro-scales, the Debye length within dielectric fluids becomes comparable to the conduit dimensions. Additionally, unique two-phase heat transport devices will be developed that will operate based on the conduction phenomenon, with or without capillary effects, in macro- and micro-scales. Heat flux levels with these active micro-scale heat transport devices are expected to significantly exceed the corresponding working fluid critical heat flux level while operating stably and for example reaching much above 300 W/cm2 for refrigerant R-123.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Innovative technology is being developed partially via this grant from NASA. Specifically, a two-phase heat transport device is being developed in meso- and micro-scale, driven with electric conduction mechanism. Two setups have been designed and fabricated based on the above mentioned concept. The first one, developed at PI's lab at IIT (to be moved to WPI in May 2012). The second one, developed as a collaborative effort between PI and Co-PI, and is located at Co-PI's lab at NASA-GSFC. The PI and Co-PI's teams are scheduled to fly on NASA's 0-g flight in April 2012 to examine this technology. This unique two-phase heat transport device actively transports heat compared to the current heat pipe technology that is a passive system.
Task Progress:	The pressure and flow generation performance of both a meso-scale and a micro-scale electrric conduction pump have been investigated. Both pumps had micro-scale electrode dimensions much less than 1 mm, representing the first time that conduction pumping has been demonstrated at these length scales. Refrigerant HCFC-123 was used as the working fluid. Both pumps were shown to produce significant pressure and flow rates. Although the efficiency of both pumps is very low, the heat transport capacity of the flow that is generated by the pump is at least two orders of magnitude larger than the input power to the pump, demonstrating that this technology is extremely viable in cooling applications where the power input to operate the cooling system is significant compared to the heat that is dissipated by the device that is being cooled.
Bibliography Type:	Description: (Last Updated: 01/30/2026)
Articles in Peer-reviewed Journals	Pearson MR, Seyed-Yagoobi J. "Experimental study of EHD conduction pumping at the meso- and micro-scale." Journal of Electrostatics. In press, September 2011. , Sep-2011
Articles in Peer-reviewed Journals	Yazdani M, Seyed-Yagoobi J. "Heat transfer augmentation of parallel flows by means of electric conduction phenomenon in macro- and micro-scales." Journal of Heat Transfer 2010 Jun;132(6):062402. http://dx.doi.org/10.1115/1.4000977 , Jun-2010
Articles in Peer-reviewed Journals	Yazdani M, Seyed-Yagoobi J. "Thermal homogenization in spherical reservoir by electrohydrodynamic conduction phenomenon." Journal of Heat Transfer, 2009 Sep;131(9):094502. http://dx.doi.org/10.1115/1.3139111 , Sep-2009
Articles in Peer-reviewed Journals	Yazdani M, Seyed-Yagoobi J. "Electrically induced dielectric liquid film flow based on electric conduction phenomenon." IEEE Transactions on Dielectrics and Electrical Insulation, 2009 Jun;16(3):768-77. http://dx.doi.org/10.1109/TDEI.2009.5128517 , Sep-2009
Dissertations and Theses	Pearson M. "Micro-Scale EHD Conduction-Driven Pumping and Heat Transfer Enhancement in Single- and Two-Phase Systems." Dissertation, Illinois Institute of Technology, Graduation Date: May 2011. , May-2011
Dissertations and Theses	Yazdani M. " Electrically Driven Liquid and Liquid Film Flow in Large and Small Scales in the Absence and Presence of Phase Change." Dissertation, Illinois Institute of Technology, Graduation Date: December 2009. , Dec-2009